HealthConnect Analytics Testing & Refinement Report

Introduction

Week 7 focused on improving the reliability, accuracy and usefulness of the analytical outputs developed during Weeks 5 and 6 of the HealthConnect project.

Week 5 established the initial exploratory data analysis, key performance indicators (KPIs), dashboard visualisations and business insights. Week 6 expanded the analysis, examined important findings and improved the dashboard.

This week's analysis concentrated on reviewing existing analytical outputs, validating KPI calculations, improving performance comparisons and identifying areas requiring further investigation.

The main objective was to ensure that HealthConnect's analytical outputs could provide reliable information to support operational monitoring and decision-making.

Testing Approach

- Important KPIs were reviewed to ensure that their calculation logic was appropriate and that the resulting values could be interpreted correctly.
- Existing findings were considered against the underlying data and the definitions used in the analysis. Particular attention was given to attendance, no-shows, appointment performance and revenue.
- The dashboard's KPI cards, date comparisons, filters and visual presentation were considered in terms of accuracy and usability.
- Existing dashboard elements were reviewed to identify opportunities to improve clarity, comparison and interpretation.
- The analysis was considered in relation to the operational decisions and improvements it could support within HealthConnect.
- KPI validation focused on ensuring that the selected indicators accurately represented the intended operational and financial measures.

Analytical Validation Results

The analytical validation framework considered:

- Whether attendance and no-show calculations use the correct appointment denominators.
- Whether cancelled and rescheduled appointments are treated consistently.
- Whether facility and department comparisons use the same definitions.
- Whether monthly comparisons use equivalent reporting periods.
- Whether differences between groups are interpreted appropriately.

The earlier analysis also highlighted the need to investigate data consistency, including possible duplicate records and discrepancies between financial measures.

These issues are important because inaccurate or inconsistent source records can affect dashboard calculations and business conclusions.

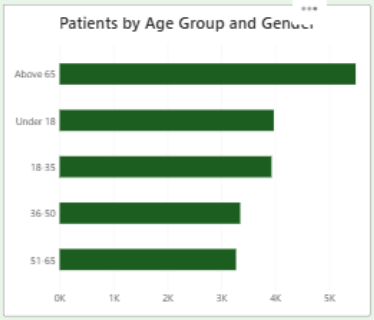
Validation status: The report documents the validation approach and the revenue comparison refinement completed during Week 7. Independent recalculation of every KPI and confirmation of all attendance/no-show findings remain subject to access to the underlying dataset.

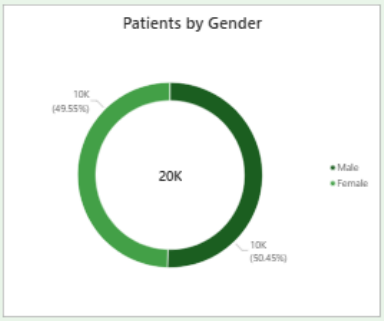
Refined dashboard/visualisations

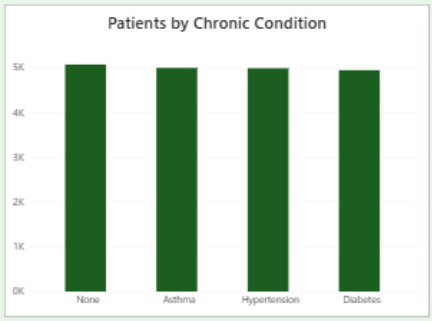


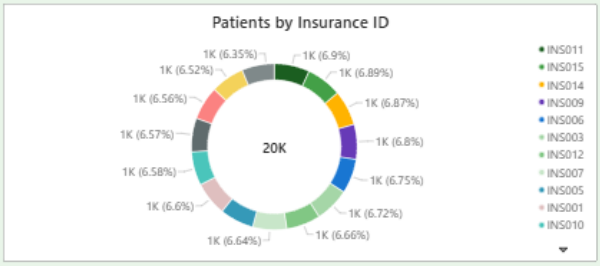
HEALTHCARE CONNECT RWANDA

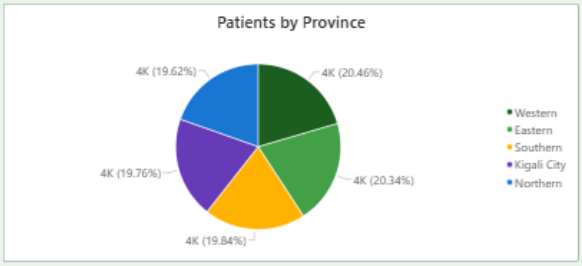
Active Patients	Total Registered Patients	Returning Patients	New Patients
20K	20K	510	4K





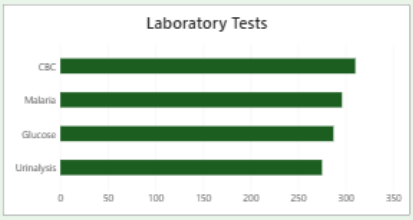


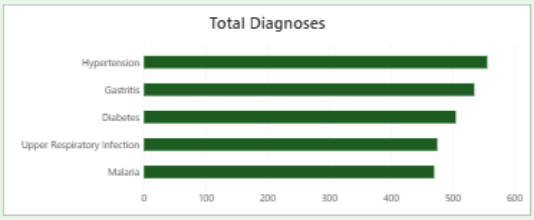


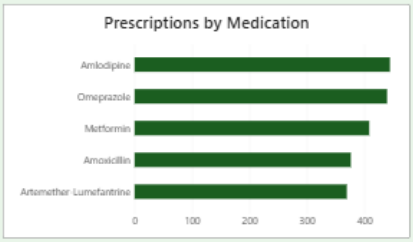


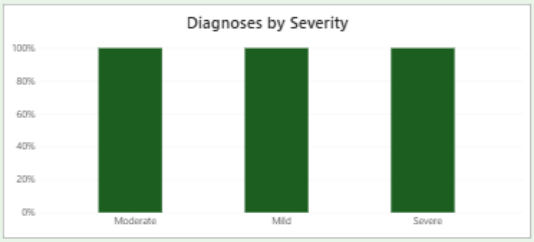
HEALTHCARE CONNECT RWANDA

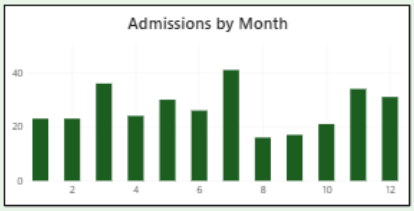
Total Admissions	Average Length of Stay	Total Appointments	Total Diagnoses
322	5.60	3K	3K

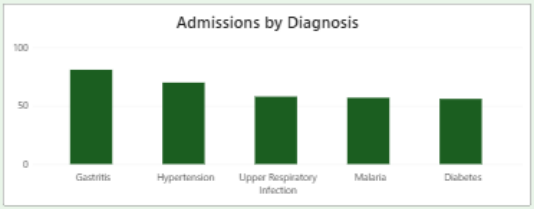






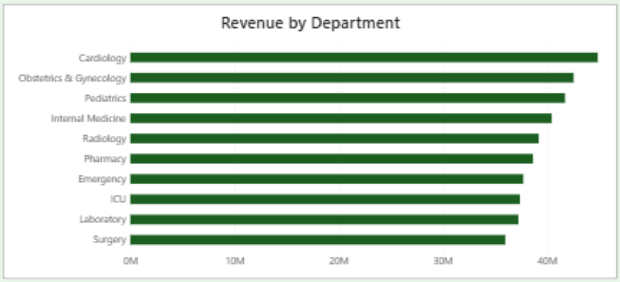


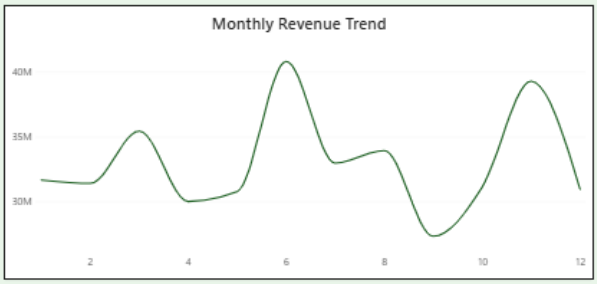


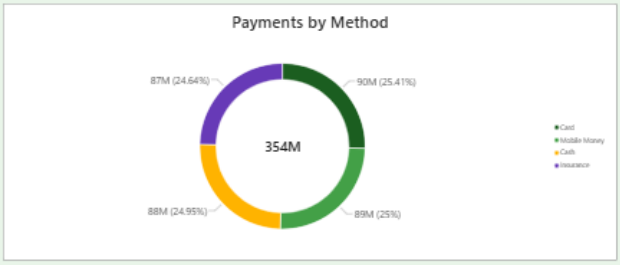


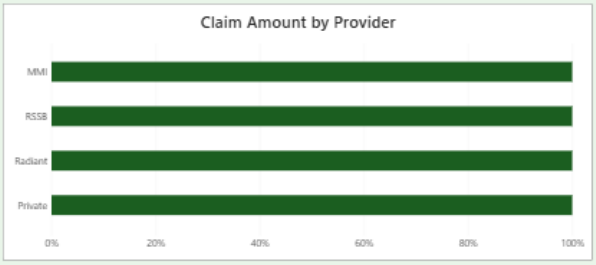
HEALTHCARE CONNECT RWANDA

Total Claims	Revenue Growth Rate	Total Insurance Claims	Total Payments	Outstanding Bills
2K	8.49%	207.37M	354M	42M









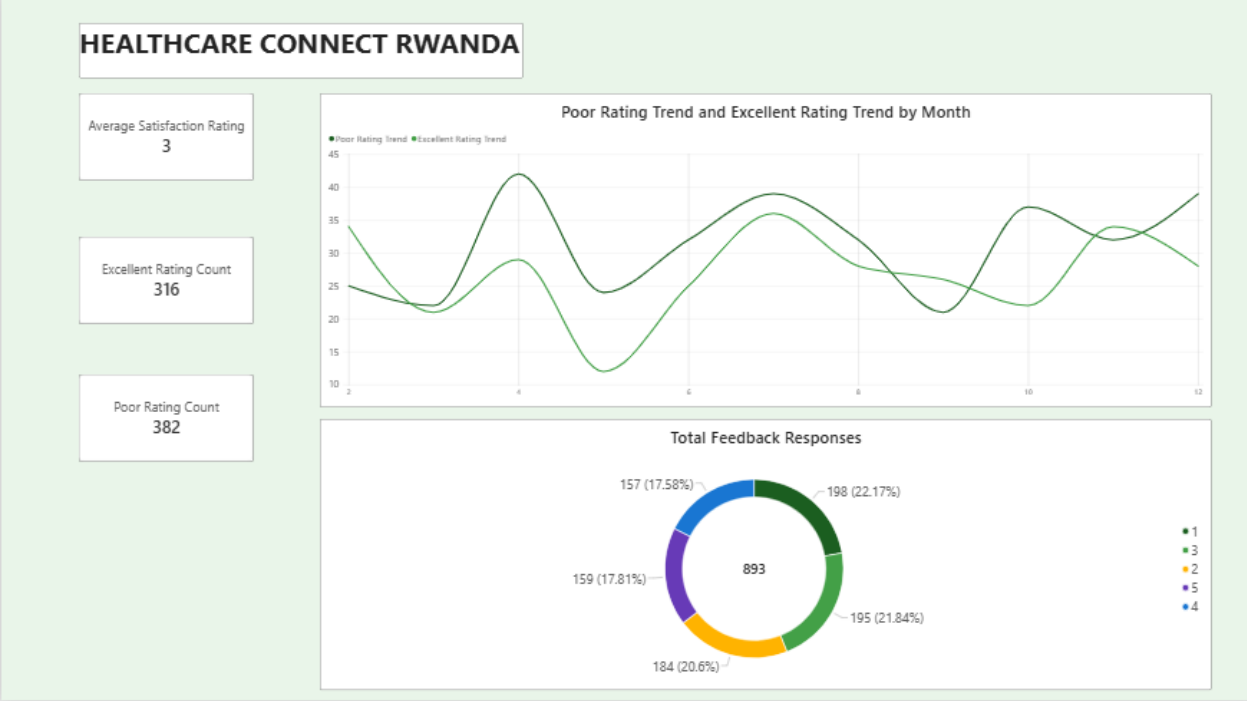
HEALTHCARE CONNECT RWANDA

Total Beds	Equipment Count	Total Rooms	Number of Radiographers
2K	200	400	30

Number of Laboratory Scientists	Number of Doctors	Number of Nurses	Number of Pharmacists
40	350	700	50

FacilityID	Total Departments	Total Revenue	Total Appointments	Total Admissions	Average Length of Stay	Total Rooms
HF004	10	22693083	143	17	5.12	20
HF009	10	22010895	135	18	5.06	20
HF012	10	21436483	138	23	6.22	20
HF019	10	21226367	125	16	6.25	20
HF008	10	21086929	134	9	4.67	20
HF014	10	20688385	135	19	6.00	20
HF006	10	20348521	123	14	4.00	20
HF018	10	20336363	139	20	6.00	20
HF017	10	20001619	124	13	5.31	20
HF011	10	19818368	121	13	4.46	20
HF007	10	19805261	130	15	5.33	20
HF010	10	19660516	131	17	5.35	20
HF001	10	19630857	122	28	5.96	20
Total	**200**	**395819453**	**2541**	**322**	**5.60**	**400**

DepartmentName	Total Appointments	Total Revenue	Total Admissions	Total Payments
Cardiology	277	44869597	40	39981110
Obstetrics & Gynecology	270	42549862	27	37875085
Pediatrics	254	41726783	29	37546535
Internal Medicine	257	40450158	26	36120844
Radiology	262	39208061	34	34996212
Pharmacy	251	38653269	43	34388118
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
Total	**2541**	**395819453**	**322**	**354103340**



Validated Key Findings

The previously reported dataset contained 20,000 registered patients, with a nearly balanced gender distribution of 10,090 males and 9,910 females. The average patient age was 45.03 years, while 5,487 patients were above the age of 65. A total of 14,929 patients were recorded as having chronic diseases, indicating a substantial chronic disease burden within the patient population. Western Province recorded the highest number of patients, with 4,091, followed closely by Eastern Province with 4,067.

Appointment analysis showed that Western Province also recorded the highest number of appointments, with 1,051, followed by Northern Province with 1,021 and Eastern Province with 1,006. Emergency appointments accounted for 1,650 of the previously reported appointments, while the average number of appointments per doctor was 14.29. These figures highlight the importance of monitoring appointment demand and healthcare staff workloads across facilities and provinces.

Clinical analysis identified gastritis, diabetes, hypertension, malaria and upper respiratory infections as the most frequently recorded diagnoses. The previously reported average length of hospital stay was 5.49 days. Laboratory testing was commonly associated with malaria, complete blood counts, glucose and urinalysis, while frequently prescribed medications included amlodipine, omeprazole and metformin. These findings suggest the need for continued monitoring of common diseases, laboratory demand and medication usage to support clinical planning.

Financial analysis previously indicated that June recorded the highest monthly revenue at approximately 70.17 million, while September recorded approximately 57.25 million. HF011 was reported as the highest-revenue facility, generating approximately 44.21 million, while Cardiology was the highest-

revenue department, with approximately 86.25 million. However, outstanding balances, insurance claims and payment figures require further reconciliation before they can be treated as fully validated financial results.

The dashboard refinement also improved revenue monitoring by introducing a previous-month comparison and a month-over-month revenue growth rate. This allows users to assess whether revenue has increased or decreased compared with the preceding month, rather than relying only on the current revenue figure. The comparison improves the interpretation of financial performance and supports more informed monitoring of revenue trends.

Updated Business Insights

The Week 7 review strengthens the interpretation of HealthConnect's analytical outputs in several ways.

Appointment management

Attendance and no-show analysis can help identify potential appointment utilisation problems. Reliable attendance indicators can support decisions about appointment reminders, scheduling and patient follow-up.

Facility and departmental performance

Facility and department comparisons can help managers identify areas requiring additional investigation. However, differences should be considered alongside appointment volumes and other relevant operational factors.

Financial monitoring

Month-over-month revenue comparison provides better visibility into changes in financial performance. Managers can identify periods of growth or decline and investigate the operational or financial factors associated with those changes.

Data reliability

The review reinforces that business decisions should rely on clearly defined KPIs and consistent source data. A visually effective dashboard is not sufficient if its underlying calculations or records are unreliable.

Updated Recommendations

Based on the Week 7 review, the following recommendations are proposed:

1. Monitor attendance and no-show rates regularly to identify potential appointment utilisation challenges.
2. Investigate facilities and departments with unusual attendance or no-show patterns.
3. Use month-over-month revenue growth to monitor changes in financial performance.
4. Investigate significant revenue increases or decreases before drawing conclusions about financial performance.
5. Maintain consistent definitions for appointments, attendance, no-shows and cancellations.
6. Review possible duplicate records and inconsistencies in the underlying dataset.
7. Ensure dashboard filters produce consistent results across relevant visualisations.
8. Continue validating KPI calculations as the dataset and dashboard evolve.

Cross-Track Testing Contribution

The analytical outputs can provide useful evidence to other HealthConnect project tracks.

HC-POD – Operations

Attendance and no-show findings can support appointment scheduling, resource planning and operational monitoring.

HC-POD – Patient Engagement

Attendance patterns can inform patient communication strategies, appointment reminders and follow-up processes.

HC-POD – Clinical Services

Department-level appointment and service patterns can help identify areas requiring further operational or clinical investigation.

HC-POD – Finance

Revenue trends and month-over-month comparisons can support financial monitoring and help identify periods requiring further investigation.

These contributions represent the potential use of the analytical findings. Findings should be shared with the relevant tracks with their validation status clearly indicated.

Limitations

The reliability of the analytical outputs depends on the quality and completeness of the underlying data.

The main limitations considered include:

- Missing or incomplete records may affect KPI calculations.
- Duplicate records may inflate counts.
- Different data-table relationships may affect aggregation.
- Attendance and no-show definitions must be consistent.
- Revenue comparisons depend on correct date relationships and complete financial records.
- Monthly changes do not necessarily explain the reasons behind performance changes.
- Observed relationships should not automatically be interpreted as causal relationships.
- Some previously reported findings have not been independently recalculated during this reporting exercise.

Recommendations

Week 8 should build on the testing and refinement work by focusing on the practical use of validated analytical outputs.

Recommended activities include:

1. Resolve outstanding data-quality and reconciliation issues.
2. Complete any remaining attendance and no-show validation.
3. Confirm dashboard filter behaviour and KPI consistency.
4. Share relevant analytical findings with the appropriate HC-POD tracks.
5. Gather feedback on the usefulness of the refined dashboard.
6. Identify opportunities to integrate validated insights into the wider HealthConnect solution.
7. Establish a process for periodically reviewing KPI accuracy.

Conclusion

Week 7 focused on strengthening the accuracy, consistency and practical usefulness of the HealthConnect analytical outputs.

The review emphasized KPI validation, attendance and no-show analysis, dashboard functionality, financial comparisons and the interpretation of business findings.

An important dashboard refinement was the introduction of month-over-month revenue growth, allowing users to compare current financial performance with the previous month.

The review also highlighted the importance of consistent data definitions, reliable source records and appropriate interpretation of analytical results.

The work provides a foundation for further validation, stakeholder engagement and integration of useful analytical findings into the wider HealthConnect solution during Week 8.